Data-Driven Color-economics:

Decoding New Growth Signals in the Electric Vehicle Sector

Final Report for ISYE 6748: Applied Analytics Practicum

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Project Objective & Business Value

The Core Problem - Seeking New Alpha in a Converging EV Market

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The Challenge & Opportunity

Beyond financial statements,
are there **quantifiable, non-financial signals** that can predict a **company's strategic maturity**?

- **The Challenge:**
  - The EV market is maturing, technology is converging, and traditional competitive advantages are eroding.
- **The Status Quo:**
  - Investment decisions require new sources of intelligence beyond traditional financial analysis.
- **Our Solution:**
  - Automotive Color, a traditionally aesthetic element, can be decoded into a unique source of "Alternative Data."

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The Solution - A Strategic Compass for VCs

Translating Color into Strategic Insight

This project delivers a data-driven analytical framework to assess the **strategic coherence, market positioning, and business model** of EV companies.

Value Proposition: To provide Qualcomm Ventures with a unique, low-cost tool to enhance due diligence and identify market opportunities.

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Methodology & Process

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Phase 1 - Quantifying "Color" into a Standardized System

From Chaos to Order

The project's foundation was built by transforming thousands of subjective color descriptions into uniform, objective data.

- **The Challenge:**
  - "Stardust Gray" vs. "Deep Space Gray"? Non-standardized names cannot be directly analyzed.
- **Our Method:**
  - **Color Space Transformation:**
    - Converted all colors from RGB to the more perceptually relevant **HSV space**.
  - **K-Means Clustering:**
    - Aggregated thousands of color points into **15 core color clusters**.
- **The Result:**
  - Established a proprietary, structured **15-dimensional color vector**, making all subsequent modeling possible.

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Phase 2 - Building a High-Performance Predictive Model

From Prediction to Interpretation

Our goal was not just to predict, but to understand "why."

- **Model:**
  - **A Multi-Output XGBoost Classifier.**
- **Key Optimizations:**
  - **Data Binarization:**
    - Focused the problem on the core business decision of "offering a color or not."
  - **Data Augmentation:**
    - Solved the real-world problem of class imbalance to ensure an unbiased model.
- **The Result:**
  - The model demonstrated excellent performance with a Jaccard Score of **0.79**, proving the strong, predictable link between vehicle parameters and color choices.

- **Jaccard Score**
  - Measures the similarity between the predicted set of colors and the actual set of colors for a vehicle.
  - It evaluates the model's accuracy in predicting the entire color portfolio. A higher score (from 0 to 1) is better.
- **Hamming Loss**
  - Calculates the fraction of color labels that are incorrectly predicted across all vehicles.
  - It provides the overall per-label error rate. A lower score (from 0 to 1) is better.

Jaccard Score	0.79
Hamming Loss	0.0646

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Core Findings

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Finding 1: The Three Tiers of Color Decisions

Who Decides the Color?

Color selection is not driven by a single factor, but by a **clear hierarchy of influences**.

- **Conclusion 1:**
  - **Brand** is the Supreme Commander.
- **Conclusion 2:**
  - **Market Segmentation** is the Battlefield.
- **Conclusion 3:**
  - **Physical & Economic Attributes** are the Foundation.

	Tier1	Tier2	Tier3
1	Manufacturer- 上汽通用五菱	Class/Category-紧凑型SUV	Energy Type-插电式混合动力
2	Manufacturer- 极氪	Class/Category-中大型MPV	Energy Type-纯电动
3	Class/Category-紧凑型SUV	Class/Category-中型车	Number of Doors
4	Manufacturer- 比亚迪	Class/Category-中大型SUV	Number of Seats
5	Class/Category-中大型MPV	Class/Category-大型SUV	Battery Fast Charge Time
6	Manufacturer- 腾势汽车	Class/Category-中大型车	First Owner Warranty
7	Manufacturer- Jeep	Energy Type-插电式混合动力	First Owner Warranty
8	Manufacturer- 零跑汽车	Energy Type-纯电动	Fuel Consumption
9	Manufacturer- 特斯拉中国	Class/Category-微型车	Max Power
10	Manufacturer- 吉利汽车	Class/Category-小型SUV	Electric Motor

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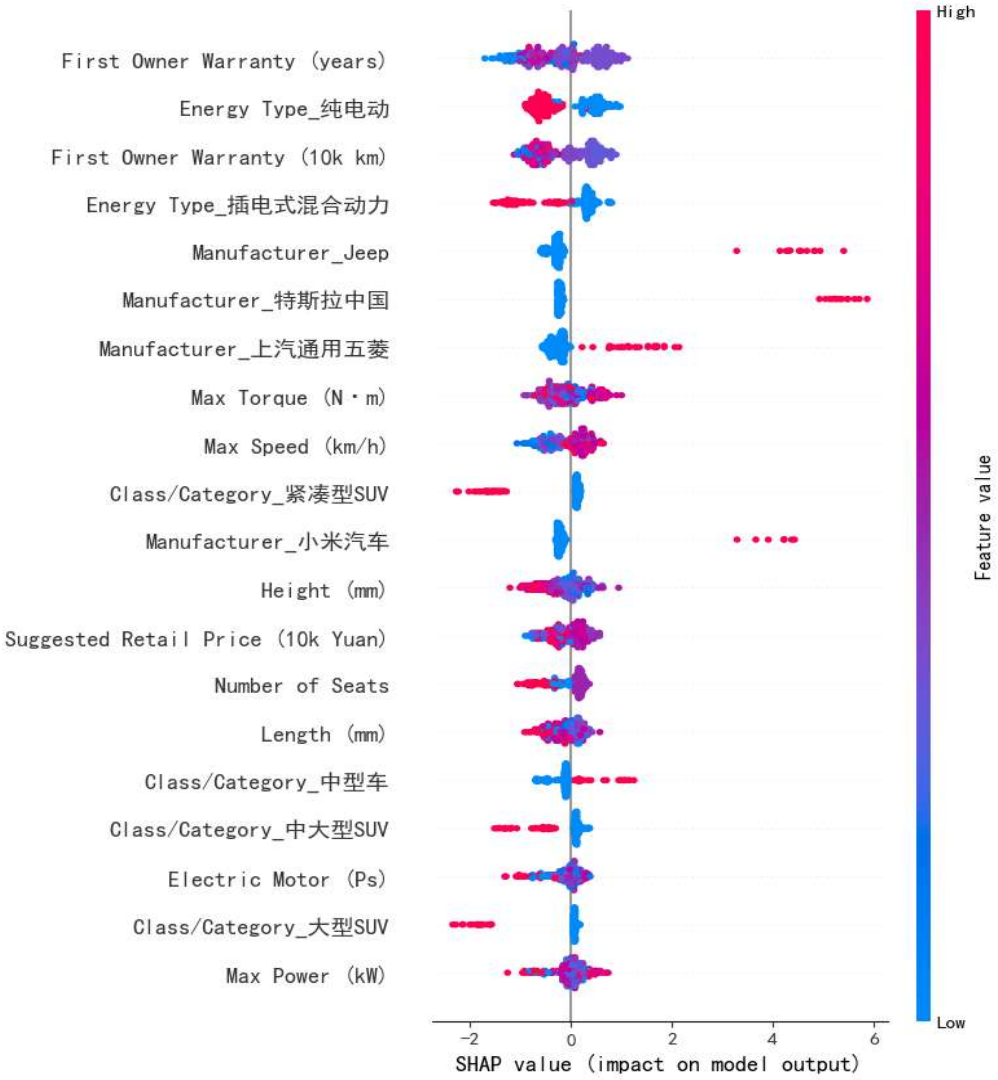
Profiling Colors - The Story of Red & Blue

Red = Performance Tag

Interpretation:

- **What the Data Says**:
  - Max Power, Horsepower, and Torque are decisive positive drivers.
  - High Warranty Requirement and Preference to High-end Manufacturer
  - Inclination to Smaller car and Higher Price
- **The Business Reality**:
  - Continuing automotive tradition, Red is the visual symbol for high performance and driving passion.
  - Red is favored by high-valuable customer who prefer sport car like ride and can afford higher premium for personality, warranty and honorable brands.

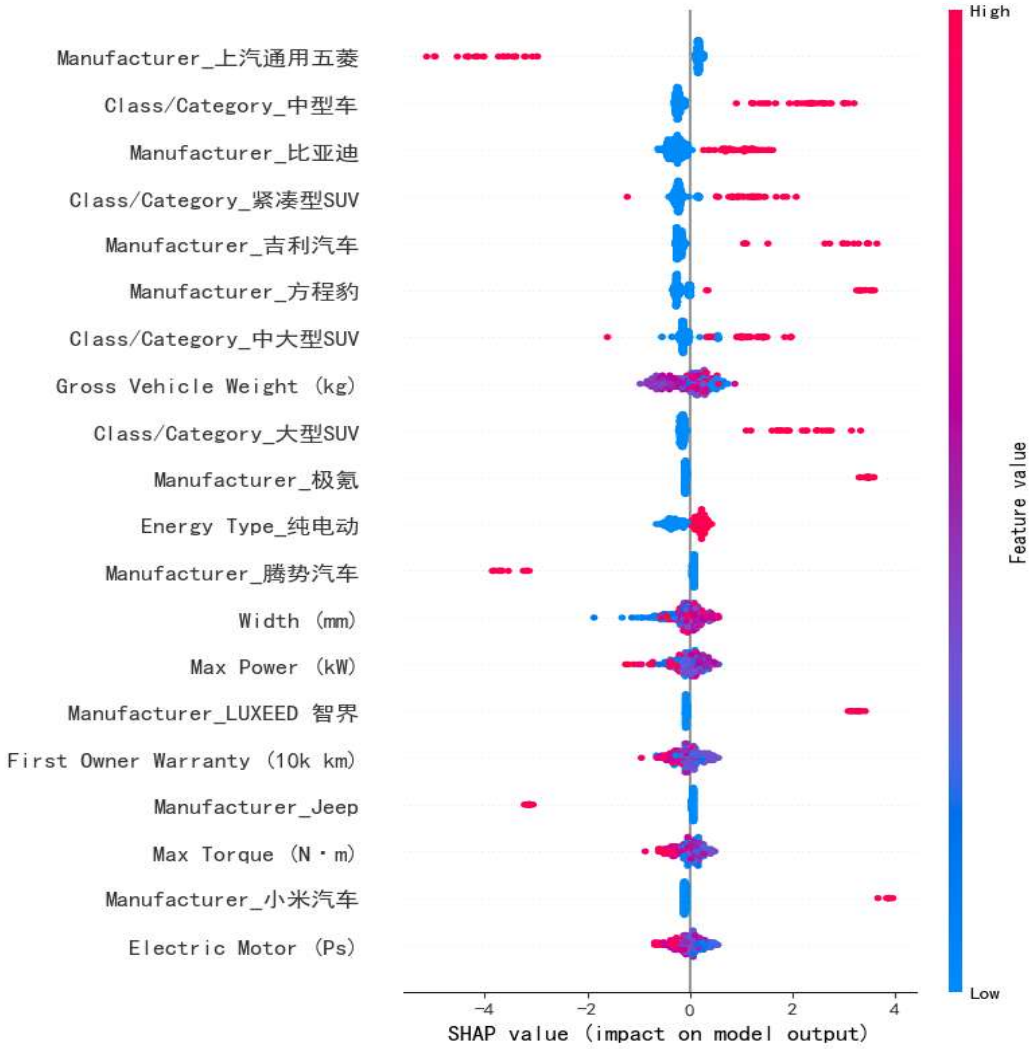


Profiling Colors - The Story of Red & Blue

Ocean Blue = Tech for the Masses

Interpretation:

- **What the Data Says**:
  - Strongly correlated with lower price and longer range.
  - Larger but Lighter Model
  - Not a fan of perfomance
- **The Business Reality**:
  - Embodied by BYD's "Ocean Series," it communicates a core value of "affordability + technology."
  - The correlated customers might be those who paying more attention to additional functions and outlooks instead of its performance, differing from vehicles' traditional criterion.

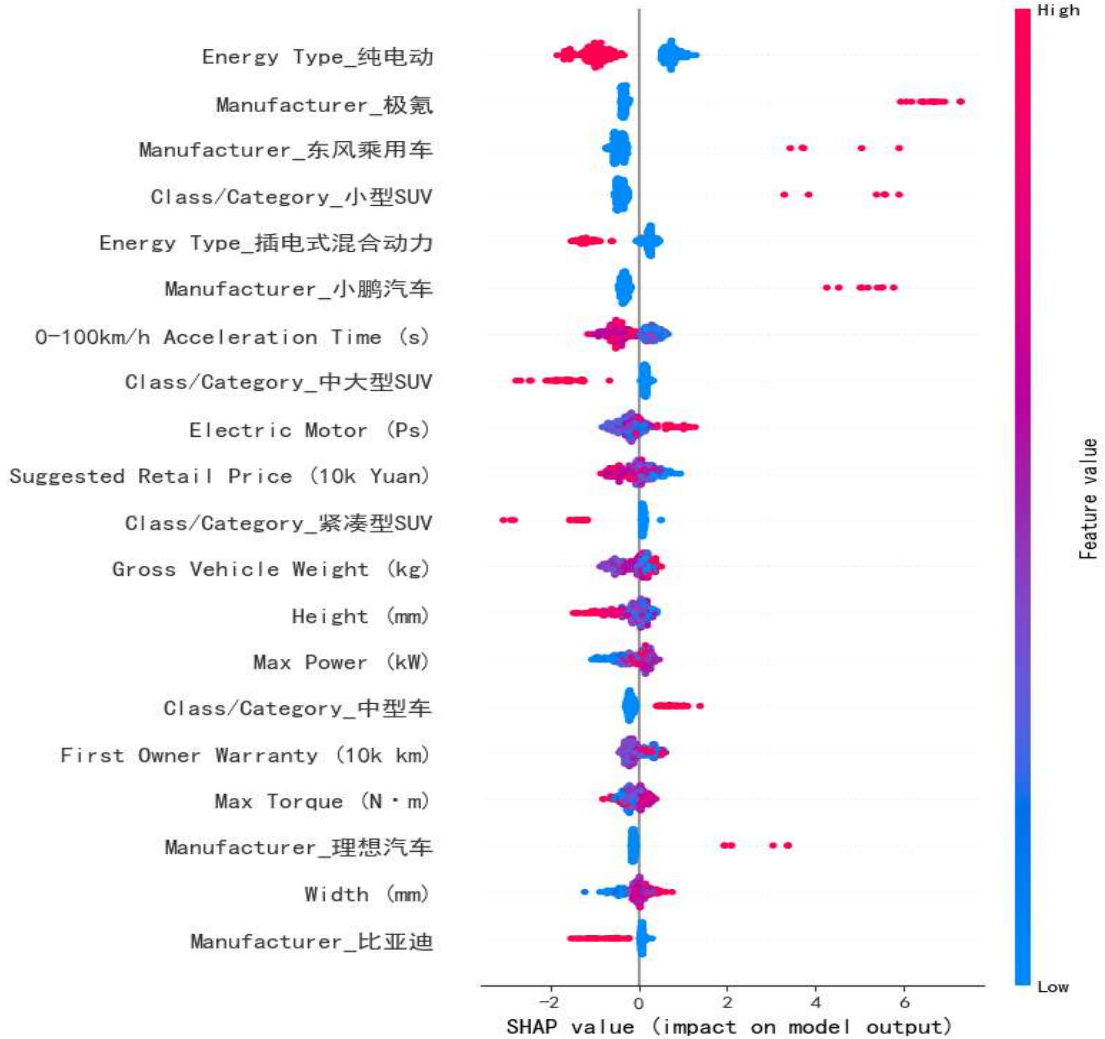


Profiling Colors – Purple: MPV Premium

Purple = The Brand Probe

Interpretation:

- **What the Data Says**:
  - There is a negative correlation with pure electric vehicles and plug-in hybrid vehicles.
  - Though often associated with a high premium, the models offering this option are usually not very expensive
- **The Business Reality**:
  - Brand Experimentation: A brand may launch a purple color option for a certain model to test market response.
  - Creating Limited Editions/Special Editions: Thanks to its uniqueness, purple is well-suited for launching special editions that can generate a price premium or buzz.
  - Adding a Personalized Aura: Introducing a purple option to an ordinary model can enhance its sense of fashion and personalization at a low cost, thereby attracting specific niche users.



The Foundational Role & Statistical Trap of Grey

Grey is the market's absolute default option. Its ubiquity presented a unique modeling challenge and confirmed midterm findings.

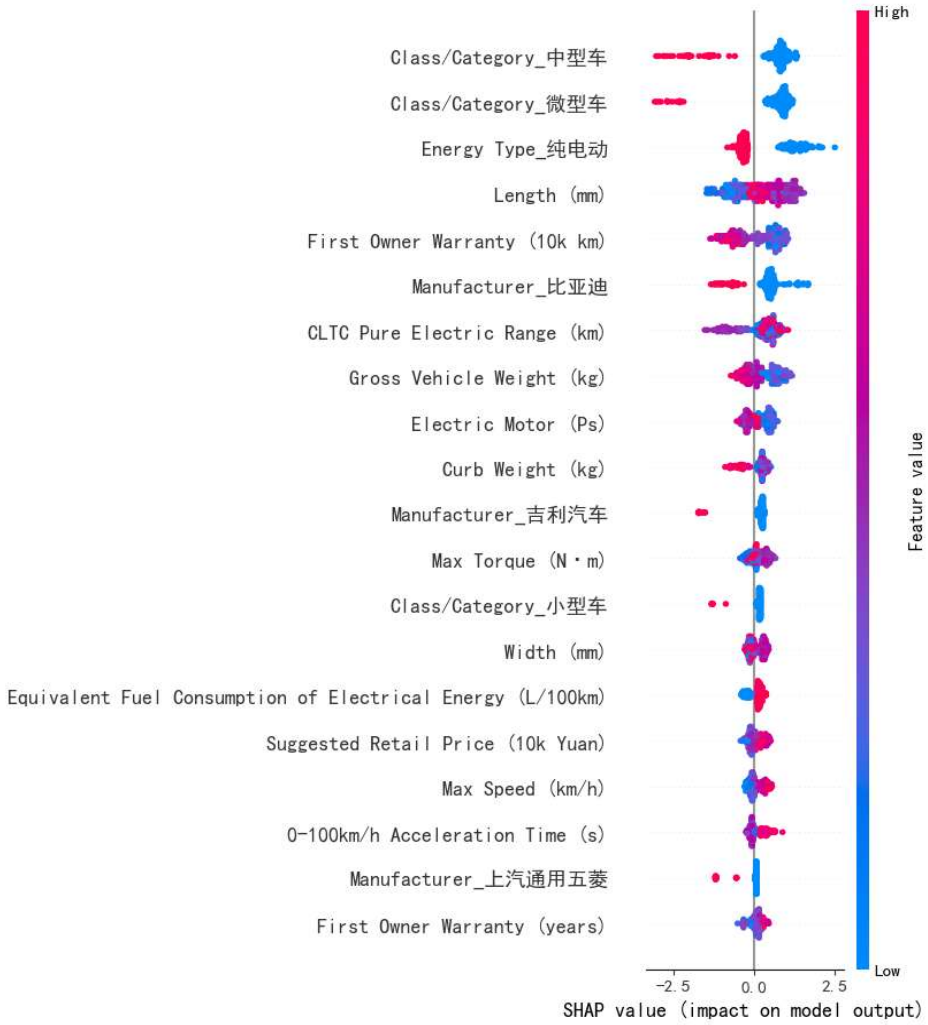
Interpretation:

- **What the Data Says**:
  - Grey is hard to predict dealing with negative examples.
  - Brands don't vary much in providing Grey.
- **The Business Reality**:
  - Dominent popularity and universal default setting.

```
--- 报告 for color: Grey ---
              precision    recall  f1-score   support

           0       0.33      0.12      0.18         8
           1       0.98      0.99      0.98       277

    accuracy                           0.97       285
   macro avg       0.65      0.56      0.58       285
weighted avg       0.96      0.97      0.96       285
```



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Investment Decision Framework

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Assessing Corporate Maturity via Color Strategy

An Actionable Investment Decision Framework

We can translate our data insights into specific questions to evaluate an investment target.

Due Diligence via a "Strategic Coherence" Test

Market Opportunity Identification via "Strategy-Mix" Analysis

Judging Ambition: "Market Follower" vs. "Market Definer"

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Case 1: Using "Green" to Vet XPeng's Strategic Risk

An Actionable Investment Decision Framework

When a company's strategy deviates from general data patterns, it constitutes a risk factor that must be questioned.

- **The Data Pattern:**
  - Our model shows that "Green" is generally associated with "economy, small size, and lower performance."
- **The Market Reality:**
  - XPeng markets its "Stardust Green" on its high-performance P7i model, attempting to link Green with "intelligent performance."
- **The Question for a VC:**
  - "Does XPeng have sufficient brand power and marketing budget to reshape the market's perception of Green?" -> This is how data insight becomes a concrete due diligence question.

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Case 2: Finding Opportunities from BYD and Wuling

Market Opportunities: Finding the Blue Ocean

Gaps between successful strategic combinations represent unmet market needs.

- **Verified Strategies:**
  - BYD: "Tech-for-the-Masses" vehicles + relatively conservative colors.
  - Wuling: "Economy Commuter" vehicles + "Macaron" fashion colors.
- **The Data-Indicated Opportunity:**
  - The market lacks players who combine these two strategies.
- **The Target for a VC:**
  - **Is there an automotive wrap/paint startup whose business model is to provide cost-effective, fashionable color upgrades specifically for "tech-for-the-masses" vehicles from BYD, Aion, etc.?** -> This is a massive market-scaling opportunity clearly indicated by the data.

Conclusion & Outlook

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Conclusion

This project successfully transformed automotive color from a subjective aesthetic into a quantifiable, interpretable business strategy.

- **Validated the Model's Efficacy:**
  - Proved that color selection is highly data-driven.
- **Revealed the Decision Hierarchy:**
  - Brand > Segmentation > Physical/Economic Attributes.
- **Decoded Color Strategies:**
  - Identified real-world business models like Performance Tagging, Tech-for-the-Masses, and the Hero Product strategy.
- **Provided a Decision Framework:**
  - Delivered a unique "Alternative Data" compass for VCs to evaluate opportunities in the EV sector.

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